Partial Differential Equations (PDE)
Assignment 1

[Transport equation]

(a) To show chain rule for composite function derivative. If $F(x) = f(g(x))$ then $F'(x) = f'(g(x))g'(x)$. If $F(t) = f(x(t), y(t))$ then $F'(t) = f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t)$. [Hint: use $f(x + \epsilon) = f(x) + f'(x)\epsilon + O(\epsilon^2)$]

(b) Use (i) the characteristic method and (ii) coordinate method to show that the solution of $au_x + bu_y = 0$ is given by $u(x, y) = f(bx - ay)$, and (iii) confirm your answer by direct substitution

(c) Solve the PDE $u_x + 2xy^2u_y = 0$ with the auxiliary condition $u = y^2$ when $x = 0$